

Exploring Predictors of Medicare Reimbursements for Blood Glucose Related Expenditures

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GitHub Repo: <https://github.com/romeor26/STATS-506>

Abstract—This study explores the socioeconomic and demographic predictors of Medicare reimbursements for blood glucose-related expenditures at the state level. Using income and tax statistics from the IRS and Medicare utilization data, we identify key factors influencing regional variations in reimbursements. Regression and Random Forest models highlight the importance of income and service density. Our findings reveal significant disparities in healthcare funding across regions, emphasizing the need for equitable access to preventive care services. By analyzing these trends, this research aims to contribute to a better understanding of resource allocation in healthcare and provide actionable insights for policymakers aiming to reduce inequities in preventive care access and funding.

I. INTRODUCTION

A. Background and Motivation

Blood glucose testing is a cornerstone of diabetes management, serving as a critical tool for both preventive and diagnostic purposes. It is particularly essential for individuals at risk of developing diabetes or managing existing conditions, as early detection and consistent monitoring can significantly reduce the risk of severe complications. Medicare, as one of the largest healthcare payers in the United States, plays a crucial role in funding these services for older adults and individuals with certain disabilities.

Despite its importance, significant disparities exist in Medicare reimbursement patterns across states, highlighting underlying socioeconomic inequalities. For example, regions with higher average incomes and service densities often experience disproportionately higher reimbursements. These patterns may exacerbate existing disparities, as lower-income regions might lack adequate funding to ensure widespread access to preventive care. Understanding the factors influencing these patterns is vital to addressing regional disparities and promoting equitable access to healthcare.

B. Project Goal

This project investigates how socioeconomic and demographic factors at the state level predict Medicare reimbursements for blood glucose-related expenditures. By integrating ZIP code-level income data from the IRS with Medicare utilization statistics, we aim to identify correlations between regional wealth, healthcare service density, and reimbursement levels. Ultimately, this analysis seeks to uncover actionable insights that can guide equitable healthcare policies, ensuring

that funding mechanisms address disparities in access to critical preventive services.

II. METHODOLOGY

A. Data Sources and LLM Used

This analysis leverages two key datasets to explore the relationship between socioeconomic factors and Medicare reimbursements:

- **IRS Data:** This dataset provides ZIP code-level statistics, including adjusted gross income (AGI), total wages, and the number of tax returns. These metrics offer a proxy for regional economic conditions and income distribution, which are critical for understanding the socioeconomic context of healthcare funding [1].
- **Medicare Data:** This dataset includes detailed information on service utilization and reimbursement amounts for various healthcare services. For this study, we filtered records to focus exclusively on blood glucose-related services, identified through relevant HCPCS codes [2].
- **ChatGPT Assistance:** ChatGPT was utilized to generate initial code suggestions, refine analysis approaches, explore statistical modeling ideas, and revise grammar issues. [3].

These datasets were merged at the state level, enabling a comprehensive analysis of how regional economic characteristics interact with healthcare reimbursement patterns.

B. Data Preparation

To prepare the datasets for analysis, the following preprocessing steps were performed:

- ****Filtering IRS Data:**** Records were filtered to include only valid AGI ranges. ZIP code-level data were aggregated to the state level, and metrics such as average income and total wages were calculated. These aggregated metrics provided a consistent basis for comparison with Medicare data.
- ****Filtering Medicare Data:**** Records were restricted to those related to glucose testing services. Invalid or missing data entries were removed, and total reimbursements were calculated by multiplying the average payment amount by the total number of services provided in each state.
- ****Merging Datasets:**** The cleaned IRS and Medicare datasets were merged by state, creating a unified dataset

that facilitated the analysis of socioeconomic predictors of Medicare reimbursements.

C. Analysis Approach

The analysis combined statistical and machine learning methods to identify key predictors of reimbursement patterns:

- **Linear Regression:** A multiple linear regression model was employed to quantify the relationships between socioeconomic predictors (e.g., average income, total services) and Medicare reimbursements. This approach provided interpretable coefficients for understanding the relative importance of each variable.
- **Random Forest:** A Random Forest model was used to capture non-linear relationships and interactions among predictors. Feature importance rankings from the model offered additional insights into which variables had the greatest impact on reimbursement patterns.
- **Geographic Mapping:** State-level variations in reimbursements were visualized using geographic maps, highlighting disparities across regions and providing a visual context for the statistical findings.

III. RESULTS

A. Geographic Disparities

Figure 1 illustrates state-level variations in total Medicare reimbursements for blood glucose-related services. Higher-income states generally report greater reimbursements, suggesting a strong correlation between regional wealth and healthcare funding. However, certain states with lower income levels received relatively high reimbursements, potentially reflecting targeted funding initiatives or inefficiencies in resource allocation.

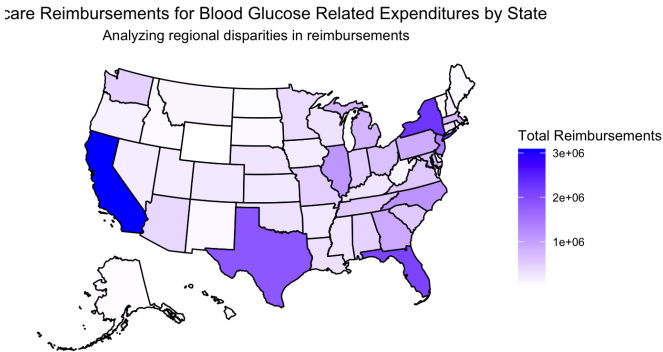


Fig. 1. State-level Medicare reimbursements for blood glucose-related expenditures.

B. Feature Importance

Results from the Random Forest model reveal that average income and total services are the most influential predictors of Medicare reimbursements (Fig. 2). These findings align closely with the regression analysis, reinforcing the importance of

regional wealth and service density in shaping reimbursement patterns. Additionally, the Random Forest model highlights subtle interactions among predictors, such as the interplay between income levels and the number of tax returns, which may not be fully captured by linear methods.

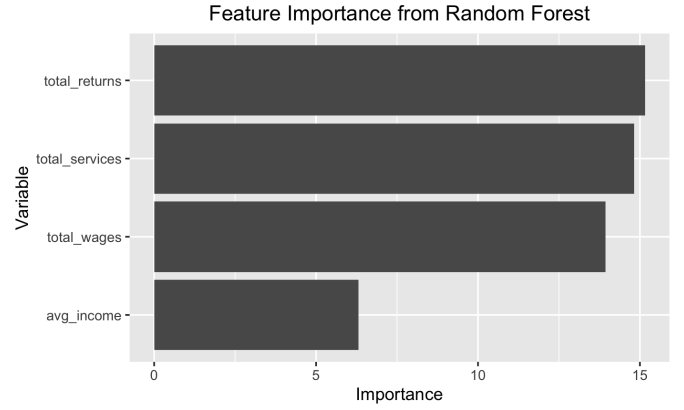


Fig. 2. Feature importance from Random Forest analysis.

IV. DISCUSSION

The findings from this analysis emphasize several critical points:

- ****Income and Service Density:**** Higher average incomes and greater service densities were consistently associated with higher Medicare reimbursements. These results suggest that wealthier regions are better positioned to attract and utilize healthcare resources, potentially leading to inequities in preventive care access.
- ****Geographic Disparities:**** While some states with lower income levels received high reimbursements, these cases were exceptions rather than the norm. The overall trend indicates that regional wealth significantly influences healthcare funding patterns.
- ****Methodological Insights:**** Combining linear regression and Random Forest models provided complementary insights, with the former offering interpretability and the latter capturing non-linear relationships and interactions among predictors.

V. CONCLUSION

This study highlights the socioeconomic factors that influence Medicare reimbursements for blood glucose-related expenditures. By identifying key predictors such as income and service density, we provide evidence of significant regional disparities in healthcare funding. These findings underscore the importance of targeted policy interventions to address inequities and ensure that preventive care services are accessible to underserved populations. Future research should explore additional factors, such as healthcare provider density and infrastructure, to gain a more comprehensive understanding of regional disparities in healthcare funding.

ATTRIBUTION OF SOURCES

REFERENCES

- [1] I. R. Service, "Soy tax stats - individual income tax statistics," <https://www.irs.gov/statistics/soi-tax-stats-individual-income-tax-statistics-2021-zip-code-data-soi-0>, accessed: December 5, 2024.
- [2] C. for Medicare & Medicaid Services, "Medicare physician & other practitioners," <https://data.cms.gov/provider-summary-by-type-of-service/medicare-physician-other-practitioners>, accessed: December 5, 2024.
- [3] ChatGPT, "Code suggestions for importing packages and random forest modeling, etc." <https://openai.com/chatgpt>, accessed: December 5, 2024.

APPENDIX

Graphs Not Included in the Main Text

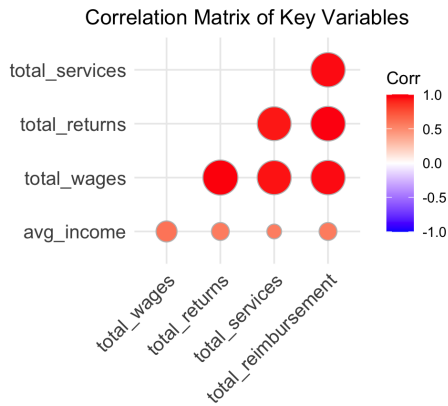


Fig. 3. Correlation matrix of key variables.

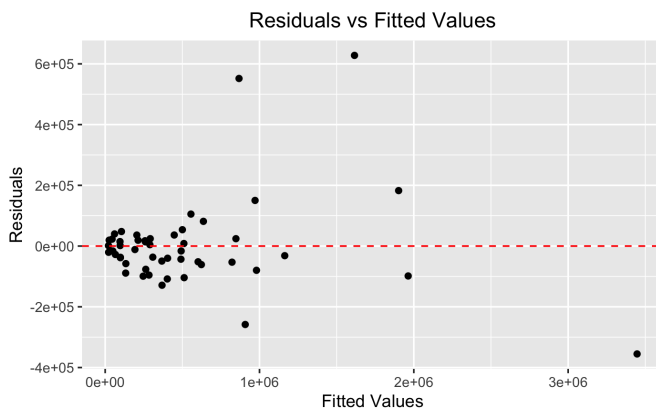


Fig. 4. Residuals vs. Fitted Values for the linear regression model.

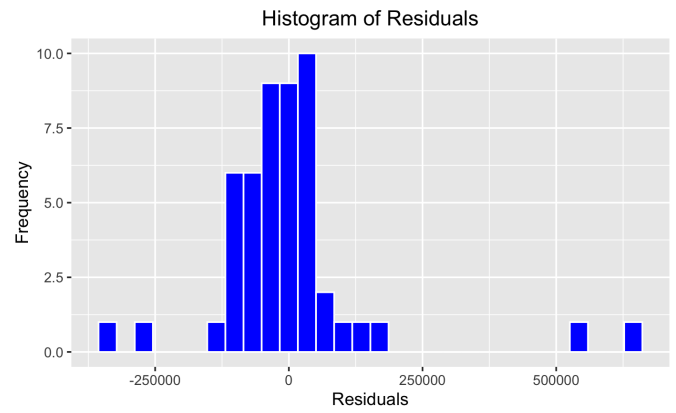


Fig. 5. Histogram of residuals for the linear regression model.

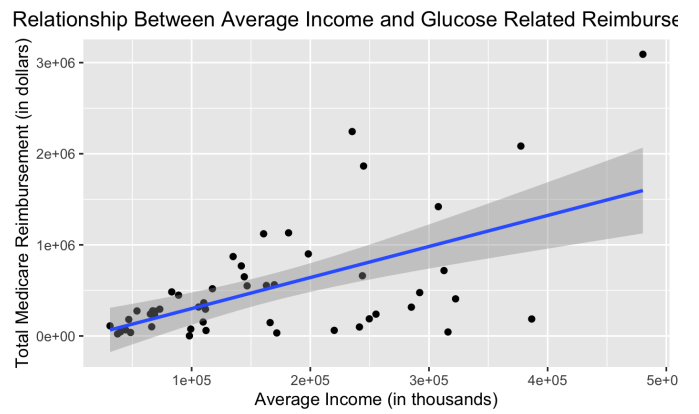


Fig. 6. Relationship Between Average Income and Glucose Related Reimbursements.

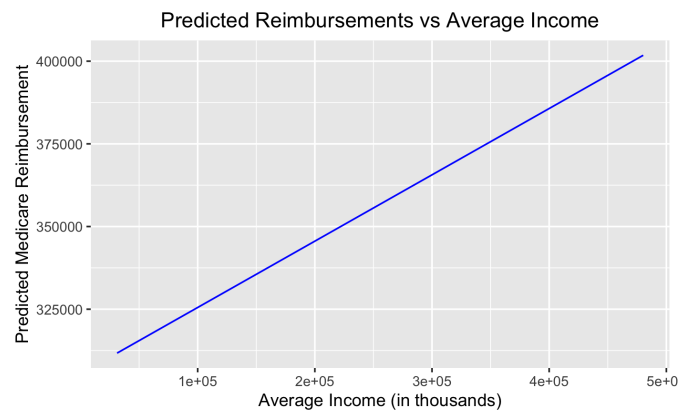


Fig. 7. Predicted Reimbursements vs Average Income.